AKSHAT MAHAJAN

Software Engineer
Indore, India | Open to relocation
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Summary

Full-stack software engineer with 3 internships shipping production web, iOS and backend systems in TypeScript, Node.js and Swift. Builds secure multi-tenant platforms end to end, from PostgreSQL schema and payment flows to real-time WebSocket interfaces. Proven record of measurable gains: 45% lower API latency, 25% faster page loads and 60% fewer redundant API calls.

Technical Skills

Languages: TypeScript, JavaScript, Python, SQL, Swift
AI & LLM: LLM APIs, RAG, structured outputs, tool calling, prompt engineering, embeddings, vector search, evaluation harnesses
Backend: Node.js, Express.js, REST APIs, WebSocket, JWT, RBAC, multi-tenancy, payment gateways (Razorpay), webhooks
Frontend: React.js, Next.js, Tailwind CSS, SwiftUI
Databases: PostgreSQL, MongoDB, Redis, Chroma (vector), Supabase (pgvector)
Infra & Testing: Docker, CI/CD (GitHub Actions), AWS S3, Jest, Supertest, structured logging, caching, rate limiting

Experience

MLock Innovations, Software Engineer Intern **September 2026 – Present**

- Architected a multi-tenant IoT self-service platform end to end: 3 Next.js/TypeScript apps on a Node.js, Express and PostgreSQL backend exposing 80+ REST endpoints
- Secured payments with server-side pricing, HMAC-verified Razorpay webhooks and idempotent capture, closing price-tampering and duplicate-charge paths across every checkout
- Built a concurrency-safe device job queue with PostgreSQL row-level locking and real-time WebSocket status, backed by JWT/RBAC tenant isolation and 200+ automated tests

DigiValet, Software Engineer Intern **January 2026 – June 2026**

- Improved application stability 25%, by reproducing intermittent runtime failures and shipping fixes from root-cause analysis and performance profiling
- Cut end-to-end API latency 45% (800ms → 440ms), by implementing Swift concurrency, request batching, and caching
- Reduced partner integration time 20% and code duplication 30%, by architecting a modular client SDK with MVVM, dependency injection, and reusable service layers

Itigai Technologies, Frontend Developer Intern **August 2025 – October 2025**

- Sustained concurrent chat sessions at under 500ms response latency, by building a real-time chatbot on WebSocket with an event-driven architecture
- Eliminated 60% of redundant API calls and cut backend load 40%, by implementing centralized state management, API caching, and request deduplication
- Reduced First Contentful Paint 25% (3.2s → 2.4s), by restructuring React.js components with memoization and code splitting

Projects

Code Journey | *Next.js, TypeScript, Supabase (pgvector), Google Gemini, Tailwind CSS*

- Launched a live career-roadmap platform that maps 10 tech roles to roughly 100 ordered skills, giving learners a clear path from beginner to job-ready
- Built CJ AI, an in-product mentor chatbot grounded in 900+ indexed passages, using hybrid vector and keyword search on pgvector with streamed, source-cited answers

Grounded RAG Document Chatbot | *Python, LLM APIs, Chroma, Ollama*

- Engineered a hallucination-resistant RAG pipeline from first principles (no LangChain): document-aware chunking, Chroma retrieval and two-layer refusal with mandatory citations
- Proved quality with a custom 14-case evaluation harness: 100% retrieval hit rate, 90% required-fact accuracy and 92.9% correct refusals on unanswerable questions

Shortify - URL Shortener | *Node.js, Express.js, PostgreSQL, Redis, Docker*

- Designed a production-style URL shortener with Base62 keys supporting 56B+ collision-free codes, and a Redis cache-aside layer that cut database load 90%
- Sustained sub-100ms redirects under load with rate limiting, request tracing and structured logging, deployed in Docker

Education

Shri Vaishnav Vidhyapeeth Vishwavidyalaya Indore, India **Aug 2022 – June 2026**
B.Tech in Computer Science Engineering